Appendix C

Background on set theory

The only thing we really need is to be aware that there's a hierarchy of sets: not every collection of things counts as a set as some are "too big". So we distinguish between *sets* and *collections*. Here "too big" means something very rigorous but quite arcane, and the definition is there to avoid Russell's paradox.

C.1 Russell's Paradox

Russell coined this paradox in 1901 and for a while it seemed like the foundations of math had fallen apart. This is because the paradox seemed to show that there was a logical contradiction at the very start, with the notion of a set. However, it turned out that this only happens if you take a very naïve definition of a set, and it can be avoided if you take a rather subtle and careful axiomatization instead.

The paradox is informally stated as "A barber shaves every man in the town who does not shave himself. Who shaves the barber?" The problem is that if the barber shaves himself then he doesn't shave himself, and if he doesn't then he does. We are stuck in a contradiction.

The paradox stems from some self-reference in the situation, and this is what happens when we state it formally as well: instead of people who may or may not shave themselves, we have sets that may or may not be an element of themselves. Remember we write "is an element of" using the symbol $\in$.

We define the following set of sets: $\boxed{\{\, S = \text{all sets } X \text{ such that } X \notin X \,\}.}$

This definition might sound confusing, and that's sort of the whole point. We then ask: is S an element of S? To untangle this more carefully, first note the following facts about any set X:

- If $X \notin X$ then $X \in S$.
- If $X \in X$ then $X \notin S$.

Now this is true for all sets X, so if we put $X = S$ we get these statements:

- If $S \notin S$ then $S \in S$.
- If $S \in S$ then $S \notin S$.

That is, either way we are doomed to a contradiction. The way to avoid it is essentially to declare that this S is not allowed to count as a set. More precisely, we set up some careful axioms for what *does* count as a set, and we are careful to avoid arbitrary collections of "all things", unless they were already elements of a set. So given something we already know is a set A, we can make a new set of "all things in A satisfying something-or-other", but we can't, out of nowhere, produce a set of "all things satisfying something-or-other". Thus, crucially, we can't produce a set of all sets, nor can we produce a set of "all sets satisfying something-or-other". Thus the thing called S in Russell's paradox no longer counts as a set, and so we can't set $X = S$ and get the contradiction.

The collection of all sets is *something*, however. We call it a "collection" and it exists at a different level up in a sort of hierarchy of sizes of collections of things. We prevent self-reference at each level, so that we don't just end up with Russell's paradox at the level above. So the totality of all collections is not a collection, but something bigger.

C.2 Implications for category theory

We have to do something analogous to avoid a Russell-like paradox in category theory. Basically we keep track of the "size" of categories and put them in a hierarchy, disallowing self-reference within any level.

So if the objects form a set, and every collection of morphisms $\mathcal{C}(a, b)$ is a set, then it's called a small category. But the totality of small categories can't be a small category: it's something one level up, which we might call "large". One level up from large we might call "super-large" and so on.

The extra subtlety with categories is that we have objects and morphisms, and even if the objects don't form a set and the *global* collection of morphisms doesn't form a set, it's possible that each individual collection of morphisms $\mathcal{C}(a, b)$ is a set. This is useful to know as we often only consider one hom-collection at a time, which is what we call doing things "locally" in a category. Categories in which every $\mathcal{C}(a, b)$ is a set are called *locally small*. They are larger than small categories but still somewhat tractable.

That's about all you really need to know to get going here. Basically bear in mind that any time we are thinking about collections of sets-with-structure, that's a type of "all sets such that [something]" so is destined to be large; however the morphisms between them are "functions such that [something]" so form a set, so the totality as a category is probably locally small.